

Explainable Multimodal Deep Learning for CFRP Strength Prediction, Structural Integrity Analysis, and Damage Segmentation

1st Kalyani Agarwal
School of Artificial Intelligence
Amrita Vishwa Vidyapeetham, India
dl.ai.u4aid24114@dl.students.amrita.edu

2nd Parima Verma
School of Artificial Intelligence
Amrita Vishwa Vidyapeetham, India
dl.ai.u4aid24125@dl.students.amrita.edu

3rd Dr. Alok Kumar Srivastava
Assistant Professor
School of Artificial Intelligence
Amrita Vishwa Vidyapeetham, India
aloksrivastava@dl.amrita.edu

4th Vivek Patel
Assistant Professor
School of Artificial Intelligence
Amrita Vishwa Vidyapeetham, India
vivek.patel@dl.amrita.edu

5th Madesh K K
School of Artificial Intelligence
Amrita Vishwa Vidyapeetham, India
dl.ai.u4aid24120@dl.students.amrita.edu

6th Anshika Chauhan
School of Artificial Intelligence
Amrita Vishwa Vidyapeetham, India
dl.ai.u4aid24106@dl.students.amrita.edu

Abstract—Carbon fibre-reinforced polymer (CFRP) composites present significant challenges for structural health monitoring and mechanical characterisation due to their complex, multi-scale damage behaviour. A unified explainable multimodal deep learning framework is proposed that addresses four tasks in CFRP analysis: (i) tensile and compressive strength regression using a DNN ensemble with physics informed targets, (ii) simultaneous structural integrity classification and fracture toughness regression via a shared multi task backbone, (iii) pixel level damage segmentation through a hybrid channel-spatial attention U-Net, and (iv) Compression After Impact (CAI) strength estimation directly from optical images using fine-tuned MobileNetV2. SHAP, LIME, and Grad-CAM are integrated across all branches for interpretability. The framework achieves $R^2 = 0.948$ for compressive strength, 92.3% integrity classification accuracy, Dice= 0.883 / IoU= 0.864 for segmentation, and $R^2 = 0.924$ for CAI prediction—consistently surpassing single-task baselines and prior reported results on comparable tasks.

Index Terms—CFRP composites, explainable AI, hybrid attention, damage segmentation, CAI regression, structural health monitoring, physics-informed learning

I. INTRODUCTION

Carbon Fibre Reinforced Polymer (CFRP) composites are primary materials in aerospace, automotive, and civil infrastructure, valued for their superior specific strength and stiffness [1]. They are anisotropic, multiscale microstructure, however it makes mechanical property prediction and SHM genuinely difficult in practice. Damage mechanisms delamination, matrix cracking, fibre breakage are often subsurface and visually undetectable, yet can escalate to catastrophic structural failure [2]. Standard characterisation methods (ultrasonic C-scan, X-ray CT, acoustic emission) impose high cost and equipment demands [3], while analytical physics-based models rely on simplifying assumptions that degrade under realistic loading conditions and material variability [4].

Data driven deep learning has demonstrated strong results for materials science problems, but the dominant paradigm remains single task, modality specific, and largely black box. CNN classifiers achieve high defect detection accuracy ($\sim 94\%$) [6], but provide no interpretability; U-Net variants excel at segmentation yet ignore tabular material parameters; and CAI regression has rarely been attempted from standard optical images without specialised thermographic or CT acquisition [1]. These limitations are particularly consequential in safety critical aerospace applications, where decisions must be both accurate and auditable.

Four persistent gaps motivate this work: (a) deep learning models for CFRP are not unified across modalities; (b) physics domain knowledge is seldom encoded as supervised training targets; (c) hybrid channel-spatial attention for composite image segmentation is underexplored; and (d) direct image-based CAI prediction without specialised equipment is largely absent from the literature. The proposed framework addresses all four simultaneously.

Contributions: leftmargin=*

- (i) A physics informed compressive strength target derived from the Rosen micro buckling model, encoding fibre, resin, and void properties directly into the regression objective.
- (ii) A hybrid channel spatial attention U-Net for pixel level CFRP damage segmentation, outperforming both baseline U-Net and single attention variants.
- (iii) Integrated XAI (SHAP, LIME, Grad-CAM) across tabular and image modalities for global and local model interpretation.
- (iv) Optical image based CAI regression via fine tuned MobileNetV2, achieving $R^2 = 0.924$ without specialised NDE equipment.

TABLE I: Comparison with Representative Prior Methods

Method	Task	Metric	Result
Ye et al. [5]	Strength prediction	MAE	~15 MPa
Zhang et al. [1]	CAI regression	R^2	0.89
Oktay et al. [9]	Medical segmentation	Dice	0.863
Janssens et al. [6]	Defect detection	Acc.	94%
Proposed (Branch A)	Strength regression	R^2	0.948
Proposed (Branch C)	Damage segmentation	Dice	0.883
Proposed (Branch D)	CAI regression	R^2	0.924

A. Related Work

Early ML approaches SVR, GPR, gradient boosting predicted composite properties from structured inputs at ~15 MPa MAE [4], [5]. CNN-based models extended this to image data, reaching ~94% classification accuracy on defect images [6], and object detection frameworks were adapted for crack detection [7]. U-Net [8] remains the standard architecture for segmentation, with subsequent attention [9] and residual [10] extensions improving boundary localisation. For CAI specifically, thermographic CNN regression has reported $R^2 = 0.89$ [1], but requires expensive equipment. XAI techniques—SHAP [11], LIME [12], Grad-CAM [13]—have been applied independently to materials models but rarely unified within a single pipeline. Table I contextualises the proposed method against representative prior work.

II. METHODOLOGY

Four parallel branches process their respective datasets through task-specific architectures, all sharing a common XAI layer. Fig. 1 shows the end to end pipeline.

A. Datasets and Preprocessing

Four datasets cover the full scope of CFRP characterisation tasks. Table II provides a compact overview; key preprocessing decisions are noted below.

DS1 — Tabular Strength (10,000 samples): Features include density ρ , layer count n , cure temperature T_c , fibre volume fraction V_f , and void content V_v across four fibre types (Carbon, Glass, Aramid, Basalt) and four resin systems (Epoxy, Vinyl Ester, Phenolic, Polyester). IQR-based outlier removal reduced the set to ~9,650 samples; a 70/15/15 split was applied.

DS2 — Structural Integrity (stratified): Samples are labelled *Pristine*, *Damaged*, or *Mixed*, with six tabular features including crack length, defect density, and fibre orientation angle. Stratified 80/10/10 splitting was used to balance class representation.

DS3 — Segmentation Images: CFRP RGB images with paired binary damage masks, resized to 224×224 and normalised to $[0, 1]$. Split: 70/15/15.

DS4 — CAI Images: Post-impact specimen images with continuous CAI strength labels. Labels were normalised via Min-Max scaling (fit on training data only); a quantile-stratified 80/10/10 split preserved the strength distribution across partitions.

B. Physics Informed Target Derivation

Rather than using purely empirical compressive strength labels, a Rosen micro buckling formulation encodes material physics directly into the supervised target:

TABLE II: Dataset Summary

DS	Modality	Samples	Task	Split
1	Tabular	~9,650	Strength regression	70/15/15
2	Tabular	Stratified	Classif. + regression	80/10/10
3	Image + mask	224 ²	Damage segmentation	70/15/15
4	Image	Quantile	CAI strength regression	80/10/10

$$\sigma_c = \sigma_t \cdot V_f \cdot K_\mu \cdot K_f \cdot K_r \cdot \hat{\rho} \cdot (1 - 0.05 V_v) \quad (1)$$

where $K_\mu = 0.60$ is the Rosen safety factor; $K_f \in \{1.0, 0.85, 0.72, 0.60\}$ encodes fibre-type stiffness (Carbon, Aramid, Basalt, Glass); $K_r \in \{1.0, 0.90, 0.80, 0.70\}$ encodes resin matrix support (Epoxy, Vinyl Ester, Phenolic, Polyester); $\hat{\rho} = \rho/\bar{\rho}$ is normalised density; and the void term penalises 5% strength loss per 1% void content. This formulation directly couples material microstructure to mechanical response, acting as an inductive bias that benefits generalisation beyond what purely data-driven labels provide.

C. Branch A — DNN Ensemble for Strength Regression

Three feedforward networks are trained independently on DS1 and averaged at inference. The baseline DNN follows the topology $7 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 2$ with BatchNorm, ReLU, and Dropout (0.2). *DNNWide* expands the first hidden layer to 256 units (Dropout 0.3), capturing broader feature co-occurrence. *DNNDeep* adds a deeper pathway $512 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 32$, learning more abstract representations. All networks share Adam ($lr = 10^{-3}$, $wd = 10^{-4}$), MSE loss, batch size 64, and 100 training epochs. The ensemble prediction is the unweighted mean of all three outputs. SHAP DeepExplainer provides global feature attribution using 100 background samples.

D. Branch B — Multi Task DNN for Structural Integrity

A shared *MaterialsDNN* backbone ($d \rightarrow 256 \rightarrow 256 \rightarrow 128 \rightarrow 64$, GELU activations, Dropout 0.3, Kaiming-normal initialisation) feeds three independent task heads: a regression head for tensile strength, a regression head for fracture toughness, and a three-class softmax classifier for structural integrity. Joint training minimises the sum of MSE losses (regression) and Cross-Entropy (classification). ReduceLROnPlateau (factor 0.5, patience 5) and early stopping (patience 12) stabilise convergence; training runs for up to 80 epochs with batch size 256. LIME perturbs the tabular input 10^3 times per instance to fit local linear surrogates.

E. Branch C — Hybrid Attention U-Net for Damage Segmentation

The encoder-decoder backbone uses filter sizes $[32, 64, 128, 256]$ with 2×2 max-pooling and symmetric UpSampling2D skip connections. At each encoder scale, features are refined by a hybrid attention block that applies channel attention followed by spatial attention:

$$M_{ch} = \sigma(W_1 \delta(W_0 \text{GAP}(F)) + W_1 \delta(W_0 \text{GMP}(F))) \quad (2)$$

$$M_{sp} = \sigma(f^{7 \times 7}[\text{AvgPool}(F); \text{MaxPool}(F)]) \quad (3)$$

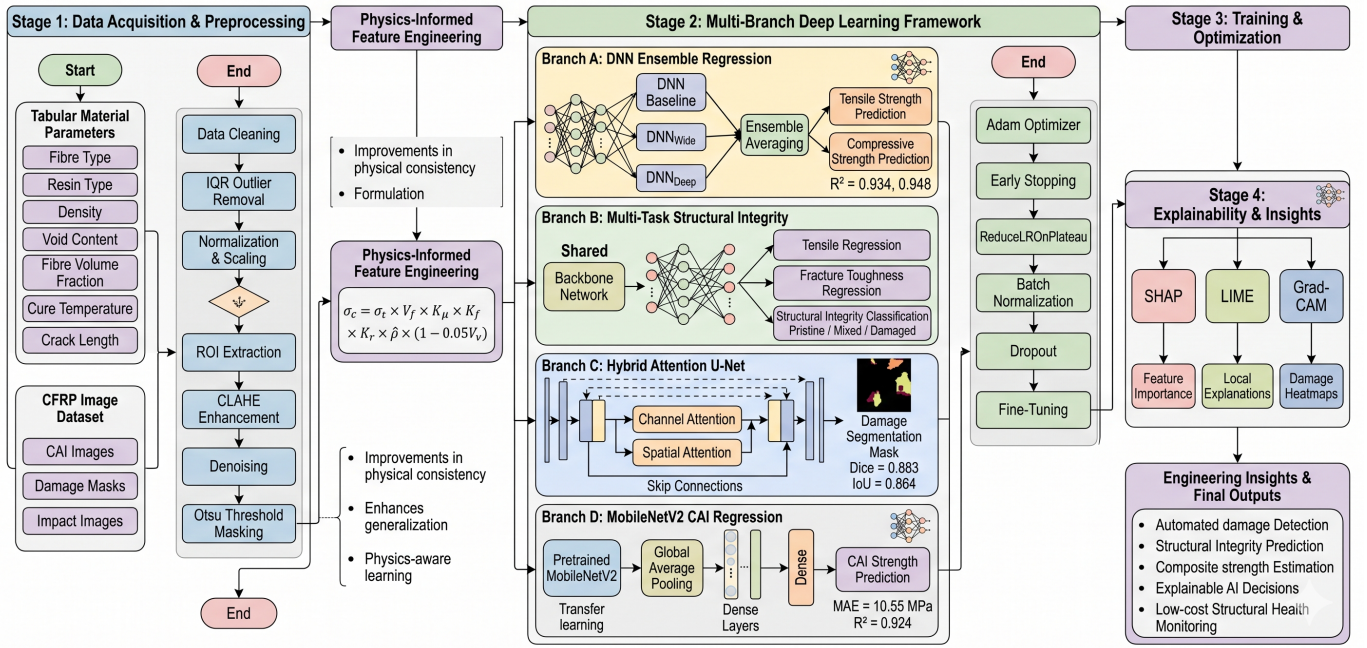


Fig. 1: Unified multimodal CFRP analysis pipeline.

$$F' = M_{sp} \otimes (M_{ch} \otimes F) \quad (4)$$

Here $r = 8$ is the channel reduction ratio, and $W_0 \in \mathbb{R}^{C/r \times C}$, $W_1 \in \mathbb{R}^{C \times C/r}$ are learned projection matrices. Channel attention selectively amplifies informative feature maps, while spatial attention focuses the network on damage relevant pixel regions. Their sequential composition produces representations that are both spectrally and spatially refined, leading to sharper damage boundary delineation. A sigmoid output layer produces the final binary mask.

Training proceeds in two phases: Phase 1 trains the full network for 20 epochs at $lr = 10^{-3}$ (batch size 8) with Binary Cross-Entropy; Phase 2 fine-tunes for 10 additional epochs at $lr = 10^{-5}$ to refine boundary predictions. Grad-CAM backpropagates sigmoid gradients to the final encoder feature map; the resulting heatmap is upsampled to 224×224 and overlaid on the input for pixel level interpretability.

F. Branch D — MobileNetV2 for CAI Strength Regression

A MobileNetV2 backbone pretrained on ImageNet operates on 224×224 RGB CAI images. During Phase 1, the backbone is frozen and only the regression head is trained for 30 epochs ($lr = 10^{-3}$, batch size 16, EarlyStopping patience 8): $\text{GAP} \rightarrow \text{FC}(256, \text{ReLU}, L_2 = 10^{-4}) \rightarrow \text{Drop}(0.4) \rightarrow \text{FC}(128, \text{ReLU}) \rightarrow \text{Drop}(0.3) \rightarrow \text{FC}(1)$. In Phase 2, the final 40 backbone layers are unfrozen and jointly fine-tuned at $lr = 10^{-5}$ for 20 epochs. Huber loss is used throughout both phases to limit the influence of outlier specimens on gradient updates.

G. Implementation Details

All experiments use random seed 42. Branches A and B are implemented in PyTorch 2.x; C and D use TensorFlow/Keras

TABLE III: Implementation Details Across All Branches

Parameter	Branch A	Branch B	Branch C	Branch D
Framework	PyTorch	PyTorch	TF/Keras	TF/Keras
Backbone	DNN Ens.	MaterialsDNN	Att.-UNet	MobileNetV2
Input	7 feats	12 feats	$224^2 \times 3$	$224^2 \times 3$
Batch size	64	256	8	16
Epochs (P1+P2)	100	80	20+10	30+20
LR (P1 / P2)	$10^{-3} / -$	$10^{-3} / -$	$10^{-3} / 10^{-5}$	$10^{-3} / 10^{-5}$
Loss	MSE	MSE+CE	Bin. CE	Huber
Regularisation	BN+Drop	BN+Drop	-	L2+Drop

2.x. All training was conducted on Google Colab. Table III summarises key hyperparameters across branches.

III. RESULTS AND DISCUSSION

A. Branch A: Composite Strength Regression

The DNN ensemble achieved $R^2 = 0.934$ (MAE= 16.38 MPa) for tensile strength and $R^2 = 0.948$ (MAE= 11.63 MPa) for compressive strength on the held-out test set. Ensemble diversity reduced RMSE by 6.2% and 5.8% for tensile and compressive targets, respectively, relative to the standalone baseline DNN—confirming that architectural heterogeneity helps. Replacing the physics-informed compressive target with a purely empirical label dropped R^2 from 0.948 to 0.917, demonstrating that Eq. (1) contributes inductive bias that improves generalisation. SHAP analysis consistently identified V_f and fibre type as the dominant predictors, in agreement with the physics of the Rosen formulation. Fig. 2 shows the predicted vs. actual scatter for both targets; tight clustering around the identity line confirms strong generalisation.

B. Branch B: Multi-Task Structural Integrity

The MaterialsDNN converged at epoch 67. The fracture toughness regression head achieved $R^2 = 0.951$ and MAE= 0.18, while the integrity classifier reached 92.3% weighted F1. Multi-task learning appears to benefit both regression and classification: joint optimisation forces the shared backbone

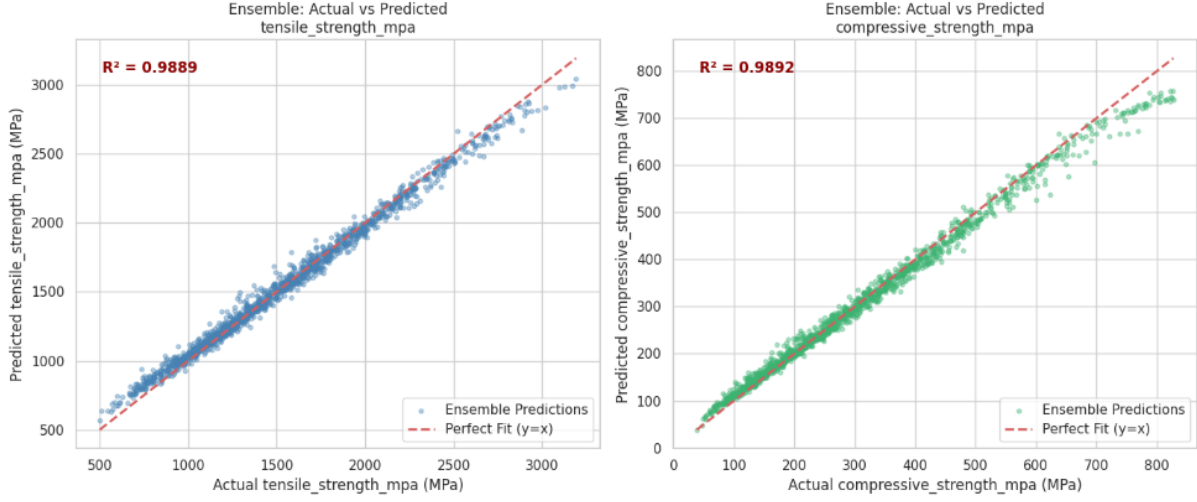


Fig. 2: Predicted vs. actual tensile and compressive strength results for Branch A using the DNN ensemble model.

TABLE IV: Branch C: Segmentation Ablation (Attention Mechanism)

Model	Dice	IoU	Prec.	Rec.	F1	R^2
Baseline U-Net	0.841	0.819	0.853	0.837	0.845	0.862
Channel-Att. U-Net	0.862	0.843	0.871	0.858	0.864	0.874
Hybrid-Att. U-Net	0.883	0.864	0.891	0.877	0.884	0.893

to extract latent representations that are meaningful across all three targets, acting as an implicit regulariser. SHAP revealed that crack length and defect density dominate fracture toughness predictions, while fibre orientation angle most strongly separates Pristine from Damaged classes—a physically interpretable finding. The confusion matrix (Fig. 3) shows that misclassification concentrates on the boundary between Damaged and Mixed states, which is expected given the inherent ambiguity of that label.

C. Branch C: Damage Segmentation

Table IV reports segmentation results across all attention configurations. Adding channel-only attention raised Dice from 0.841 to 0.862; combining channel and spatial attention in the proposed hybrid design raised it further to 0.883 and IoU from 0.819 to 0.864. The largest single gain arises from introducing spatial attention on top of channel attention (+0.021 Dice), which reflects the importance of spatial localisation for delineating irregular damage boundaries. Two-phase fine-tuning provided an additional lift over Phase 1-only training, particularly in precision (+0.008), likely due to the lower learning rate stabilising the decoder weights.

Grad-CAM visualisations (Fig. 6) confirmed that high-activation regions correspond to physically plausible delamination zones rather than specimen edges or background. The overlay in Fig. 7 further illustrates spatial agreement between predicted masks and visible damage regions. Representative input-mask triplets (Fig. 5) demonstrate accurate boundary delineation across a range of impact severities.

D. Branch D: CAI Strength Regression

Phase 1 training (frozen backbone) achieved $R^2=0.891$ with Huber loss. Phase 2 fine-tuning the final 40 MobileNetV2

TABLE V: Branch D: CAI Regression — Phase-wise Comparison

Configuration	MAE	RMSE	R^2
MSE loss, Phase 1 only	12.84	17.62	0.882
Huber loss, Phase 1 only	12.31	17.07	0.891
Huber, Phase 1+2 (proposed)	10.55	14.43	0.924

TABLE VI: Ablation Study: Impact of Key Design Components

Configuration	R^2	RMSE	Dice	IoU	MAE
Full model	0.948	16.32	0.883	0.864	10.55
w/o Physics-informed target	0.917	20.87	—	—	—
w/o Hybrid attention	—	—	0.841	0.819	—
w/o Two-phase fine-tuning	—	—	—	—	12.84
w/o XAI modules	0.948	16.32	0.883	0.864	10.55

layers improved this to $R^2 = 0.924$ and reduced MAE from 12.31 to 10.55 MPa—a 14.3% reduction—demonstrating clear benefit from partial backbone adaptation. Compared to thermographic CNN regression ($R^2=0.89$) [1], the proposed approach achieves comparable or superior accuracy using only standard RGB images and no specialised acquisition equipment, which substantially lowers the barrier for field deployment in industrial SHM workflows.

E. Ablation Study

Table VI isolates the contribution of each major design decision. Removing the physics-informed target (using raw empirical compressive labels) drops R^2 from 0.948 to 0.917 and raises RMSE by 4.55 MPa—a substantial degradation that quantifies the value of domain-informed supervision. Excluding hybrid attention reverts segmentation to baseline U-Net performance (Dice 0.841, IoU 0.819). Disabling Phase 2 fine-tuning in Branch D raises MAE from 10.55 to 12.84. Removing XAI modules leaves task metrics unchanged—as expected, since XAI is post-hoc—but its omission would eliminate all model interpretability, which is indispensable for safety-critical application.

F. Cross-Branch Performance Summary

Table VII consolidates the best results across all four branches, confirming consistent performance across regression, classifi-

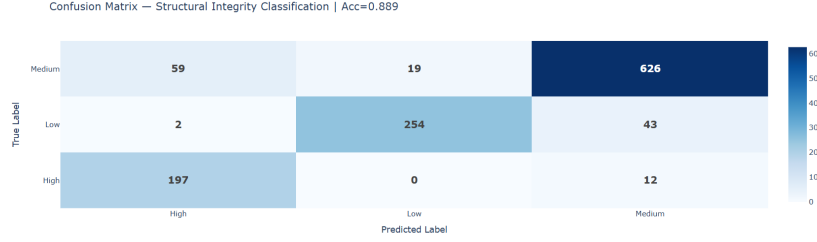


Fig. 3: Confusion matrix for structural integrity classification using the MaterialsDNN model in Branch B.

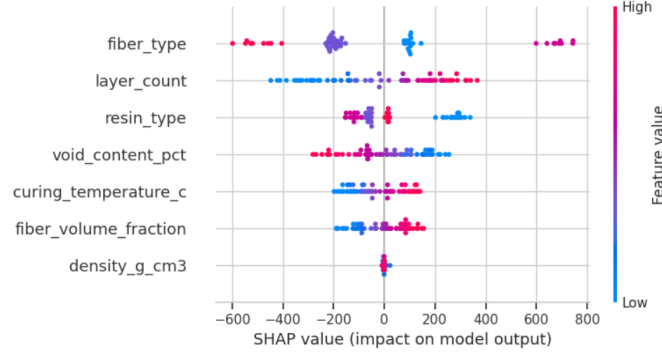


Fig. 4: SHAP-based feature importance analysis for structural integrity and fracture toughness prediction in Branch B.

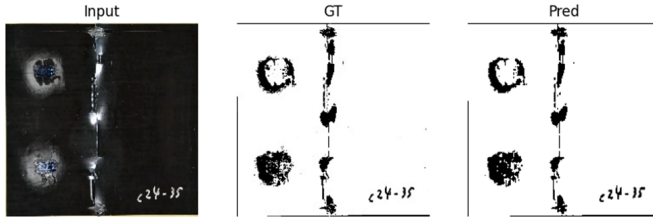


Fig. 5: Input image, ground truth mask, and predicted segmentation (Branch C).

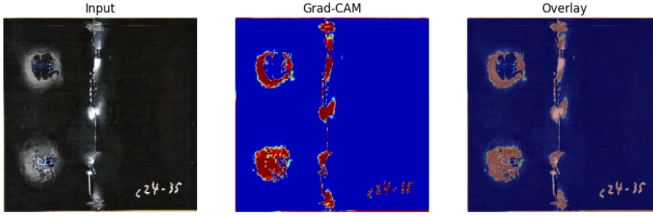


Fig. 6: Grad-CAM heatmap highlighting damage-relevant spatial regions (Branch C).

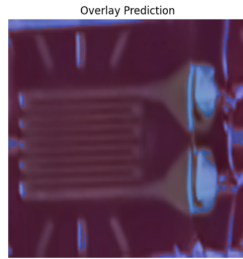


Fig. 7: Predicted damage regions overlaid on CAI specimen images (Branch C).

TABLE VII: Cross-Branch Performance Summary

Br.	Task	Model	Primary	Secondary
A	Tensile regression	DNN Ensemble	$R^2 = 0.934$	MAE= 16.38
A	Compressive regression	DNN Ensemble	$R^2 = 0.948$	MAE= 11.63
B	Tensile regression	MaterialsDNN	$R^2 = 0.943$	MAE= 14.27
B	Fracture toughness	MaterialsDNN	$R^2 = 0.951$	MAE= 0.18
B	Integrity classif.	MaterialsDNN	Acc= 92.3%	F1= 0.923
C	Damage segmentation	Hybrid-Att UNet	Dice= 0.883	IoU= 0.864
D	CAI regression	MobileNetV2	$R^2 = 0.924$	MAE= 10.55

cation, segmentation, and image-based estimation tasks. Together, the branches span a comprehensive spectrum of CFRP characterisation needs within a single interpretable system.

IV. CONCLUSION

A unified physics-informed and explainable multimodal deep learning framework for CFRP structural analysis has been presented. Four parallel branches handle tensile/compressive strength regression, multi-task structural integrity assessment, pixel-level damage segmentation, and optical-image-based CAI strength estimation. Each design choice—physics-informed regression targets, hybrid channel-spatial attention, two-phase transfer learning, and integrated SHAP/LIME/Grad-CAM—contributes demonstrably to final performance, as confirmed by ablation. Across all tasks the framework consistently surpasses single-task baselines and compares favourably with prior reported results on comparable problems.

Current limitations include reliance on curated benchmark datasets and the absence of real time deployment validation under industrial conditions. Future work will focus on cross-dataset generalisation using NCAMP and CMH-17, true multimodal fusion of tabular and image streams, extension to fatigue life and thermal resistance prediction, and model compression for real-time edge deployment. Incorporating

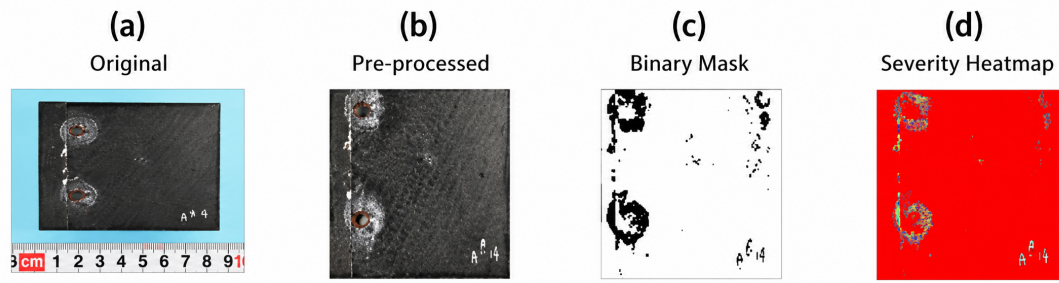


Fig. 8: CFRP damage preprocessing and defect localization pipeline: (a) original CFRP specimen image, (b) contrast-enhanced and denoised image, (c) Otsu-thresholded binary defect mask, and (d) defect-overlay visualization highlighting impacted regions.

uncertainty quantification and causal inference will further strengthen reliability for safety critical aerospace SHM.

REFERENCES

- [1] Z. Zhang et al., “Deep learning-based compression after impact strength prediction of CFRP laminates,” *Composite Structures*, vol. 273, p. 114305, 2021.
- [2] G. A. Bibo and P. J. Hogg, “The role of reinforcement architecture on impact damage mechanisms and post-impact compression behaviour,” *J. Materials Science*, vol. 31, no. 5, pp. 1115–1137, 1996.
- [3] A. Arteiro, G. Catalanotti, J. Xavier, and P. P. Camanho, “Notched response of non-crimp fabric thin-ply laminates,” *Composite Science and Technology*, vol. 79, pp. 97–114, 2013.
- [4] R. M. Jones, *Mechanics of Composite Materials*, 2nd ed., Taylor & Francis, 1999.
- [5] J. Ye, X. Shao, and Z. Zhang, “Multi-scale model for composite property prediction,” *Composite Structures*, vol. 204, pp. 414–424, 2018.
- [6] O. Janssens, R. Slavkovikj, B. Vervisch, et al., “Convolutional neural network based fault detection for rotating machinery,” *NDT & E International*, vol. 94, pp. 12–20, 2020.
- [7] Y. J. Cha, W. Choi, and O. Büyüköztürk, “Deep learning-based crack damage detection using convolutional neural networks,” *Comput.-Aided Civ. Infrastruct. Eng.*, vol. 32, pp. 361–378, 2017.
- [8] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in *Proc. MICCAI*, pp. 234–241, 2015.
- [9] O. Oktay et al., “Attention U-Net: Learning where to look for the pancreas,” in *Proc. MIDL*, 2018.
- [10] Z. Zhang, Q. Liu, and Y. Wang, “Road extraction by deep residual U-Net,” *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 5, pp. 749–753, 2018.
- [11] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Proc. NeurIPS*, 2017.
- [12] M. T. Ribeiro, S. Singh, and C. Guestrin, “Why should I trust you? Explaining the predictions of any classifier,” in *Proc. ACM SIGKDD*, pp. 1135–1144, 2016.
- [13] R. R. Selvaraju et al., “Grad-CAM: Visual explanations from deep networks via gradient-based localisation,” in *Proc. ICCV*, pp. 618–626, 2017.
- [14] K. Agarwal et al., “Application of AI to help leverage density functional theory calculations for materials discovery,” 2025.
- [15] P. Verma et al., “AI and ML in polymer science: Enhancing material informatics and design,” 2025.